

Supplementary material of the bioacoustics analyses

Table 1. Recordings analyzed and the groups delimitation used. Acronyms of the voucher codes: Arquivo Sonoro Elias Coelho (ASEC); Fonoteca Neotropical Jacques Vielliard (FNJV); Macaulay Library (ML); WikiAves (WA); Xeno-canto (XC).

Voucher	Country	Locality	Group
ML87660	Bolivia	Cochabamba	<i>boliviana</i>
ML101625	Bolivia	La Paz	<i>boliviana</i>
ML132722	Bolivia	La Paz	<i>boliviana</i>
ML120944	Bolivia	La Paz	<i>boliviana</i>
ML17831	Bolivia	La Paz	<i>boliviana</i>
ML404565891	Bolivia	Santa Cruz	<i>boliviana</i>
ML54882001	Brazil	Bahia	<i>campanisona</i>
ML257218281	Brazil	Bahia	<i>campanisona</i>
ML391926281	Brazil	Bahia	<i>campanisona</i>
ML320720641	Brazil	Bahia	<i>campanisona</i>
ML59451421	Brazil	Bahia	<i>campanisona</i>
ML627903020	Brazil	Bahia	<i>campanisona</i>
ASEC10666	Brazil	Bahia	<i>campanisona</i>
ASEC10667	Brazil	Bahia	<i>campanisona</i>
ML94935	Brazil	Espírito Santo	<i>campanisona</i>
ML238852	Brazil	Espírito Santo	<i>campanisona</i>
ML425734951	Brazil	Espírito Santo	<i>campanisona</i>
WA5732556	Brazil	Minas Gerais	<i>campanisona</i>
ML362446751	Brazil	Paraná	<i>campanisona</i>
ML174392	Brazil	Paraná	<i>campanisona</i>
ML608981043	Brazil	Paraná	<i>campanisona</i>
FNJV29625	Brazil	Paraná	<i>campanisona</i>
ASEC11971	Brazil	Paraná	<i>campanisona</i>
FNJV103438	Brazil	Rio de Janeiro	<i>campanisona</i>
XC287752	Brazil	Rio de Janeiro	<i>campanisona</i>
XC386878	Brazil	Rio de Janeiro	<i>campanisona</i>
XC726902	Brazil	Rio de Janeiro	<i>campanisona</i>
ASEC12872	Brazil	Rio de Janeiro	<i>campanisona</i>
ASEC1196	Brazil	Rio de Janeiro	<i>campanisona</i>
ML626340742	Brazil	Santa Catarina	<i>campanisona</i>
ML626435624	Brazil	Santa Catarina	<i>campanisona</i>
ML21274201	Brazil	Santa Catarina	<i>campanisona</i>
ML94925	Brazil	São Paulo	<i>campanisona</i>
FNJV13774	Brazil	São Paulo	<i>campanisona</i>
FNJV14836	Brazil	São Paulo	<i>campanisona</i>
XC770850	Brazil	São Paulo	<i>campanisona</i>
XC424636	Brazil	São Paulo	<i>campanisona</i>
XC80408	Brazil	São Paulo	<i>campanisona</i>
ML430243231	Brazil	São Paulo	<i>campanisona</i>
ML451939891	Brazil	São Paulo	<i>campanisona</i>
ML233117361	Colombia	Boyacá	<i>columbiana</i>
ML608746673	Colombia	Caquetá	<i>columbiana</i>

Continuation of the Table 1.

Voucher	Country	Locality	Group
ML89353	Colombia	Caquetá	<i>columbiana</i>
ML619037508	Colombia	Casanare	<i>columbiana</i>
ML635581411	Colombia	Casanare	<i>columbiana</i>
ML254025	Colombia	Nariño	<i>columbiana</i>
ML254023	Colombia	Nariño	<i>columbiana</i>
ML23611931	Colombia	Putumayo	<i>columbiana</i>
XC917331	Colombia	Santa Rosa del Sur	<i>columbiana</i>
ML570584681	Brazil	Roraima	<i>fulvescens</i>
ML195649	Guyana	Mt. Ayanganna	<i>fulvescens</i>
ML195656	Guyana	Mt. Ayanganna	<i>fulvescens</i>
ML134235	Guyana	Potaro-Siparuni	<i>fulvescens</i>
FNJV8584	Brazil	Alagoas	North SFR
ML235964	Brazil	Ceará	North SFR
ML231702021	Brazil	Ceará	North SFR
ML621540303	Brazil	Ceará	North SFR
FNJV3027	Brazil	Ceará	North SFR
FNJV16021	Brazil	Ceará	North SFR
ML540939651	Brazil	Ceará	North SFR
FNJV36311	Brazil	Ceará	North SFR
ML305981161	Venezuela	Bolívar	<i>obscura</i>
ML303156961	Venezuela	Bolívar	<i>obscura</i>
ML300104351	Venezuela	Bolívar	<i>obscura</i>
ML300104361	Venezuela	Bolívar	<i>obscura</i>
ML388976641	Venezuela	Bolívar	<i>obscura</i>
ML388976711	Venezuela	Bolívar	<i>obscura</i>
ML147463	Peru	Cusco	<i>olivacea/berlepschi</i>
ML286563021	Peru	Cuzco	<i>olivacea/berlepschi</i>
ML611622290	Peru	Huánuco	<i>olivacea/berlepschi</i>
ML17797	Peru	Huánuco	<i>olivacea/berlepschi</i>
ML98864	Peru	Pasco	<i>olivacea/berlepschi</i>
ML65737651	Peru	Puno	<i>olivacea/berlepschi</i>
ML31740731	Peru	Puno	<i>olivacea/berlepschi</i>
ML66442411	Peru	Puno	<i>olivacea/berlepschi</i>
ML284527191	Ecuador	Morona-Santiago	<i>punctigula</i>
ML78538	Ecuador	Napo	<i>punctigula</i>
ML242590	Ecuador	Napo	<i>punctigula</i>
ML243020	Ecuador	Napo	<i>punctigula</i>
ML219926591	Ecuador	Napo	<i>punctigula</i>
ML618908682	Ecuador	Zamora-Chinchipe	<i>punctigula</i>
ML318535091	Ecuador	Zamora-Chinchipe	<i>punctigula</i>
ML77201181	Ecuador	Zamora-Chinchipe	<i>punctigula</i>
ML225154101	Peru	Junín	<i>punctigula</i>
ML608677863	Peru	San Martín	<i>punctigula</i>
ML43558421	Peru	San Martín	<i>punctigula</i>
ML166327	Peru	Ucayali	<i>punctigula</i>
ML138817	Peru	Ucayali	<i>punctigula</i>
ML132494	Argentina	Misiones	<i>tshororo</i>
ML50486	Argentina	Misiones	<i>tshororo</i>
ML330732741	Argentina	Misiones	<i>tshororo</i>

Continuation of the Table 1.

Voucher	Country	Locality	Group
ML143620	Argentina	Misiones	<i>tshororo</i>
ML613565109	Argentina	Misiones	<i>tshororo</i>
FNJV3029	Brazil	Mato Grosso do Sul	<i>tshororo</i>
ML182077421	Brazil	Paraná	<i>tshororo</i>
ML192117581	Brazil	Paraná	<i>tshororo</i>
ML618522681	Brazil	Paraná	<i>tshororo</i>
ML238872	Brazil	Paraná	<i>tshororo</i>
ML19275	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML19273	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML25438	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML108359291	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML216492561	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML19274	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML481915541	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML19291	Brazil	Rio Grande do Sul	<i>tshororo</i>
ML535244221	Brazil	Santa Catarina	<i>tshororo</i>
ML143532	Paraguay	Caazapá	<i>tshororo</i>
ML143515	Paraguay	Caazapá	<i>tshororo</i>
ML54342	Paraguay	Caazapá	<i>tshororo</i>
ML40427	Venezuela	Aragua	<i>venezuelana</i>
ML53665	Venezuela	Aragua	<i>venezuelana</i>
ML390586041	Venezuela	Aragua	<i>venezuelana</i>
ML383315281	Venezuela	Aragua	<i>venezuelana</i>
ML62153	Venezuela	Aragua	<i>venezuelana</i>
ML62151	Venezuela	Aragua	<i>venezuelana</i>
ML386695291	Venezuela	Barinas	<i>venezuelana</i>
ML626501448	Venezuela	Carabobo	<i>venezuelana</i>
ML287002	Venezuela	Carabobo	<i>venezuelana</i>
ML606702681	Venezuela	Distrito Capital	<i>venezuelana</i>
ML336721141	Venezuela	Distrito Capital	<i>venezuelana</i>
ML617627896	Venezuela	Distrito Capital	<i>venezuelana</i>
ML617983453	Venezuela	Distrito Capital	<i>venezuelana</i>
ML574970271	Venezuela	Distrito Capital	<i>venezuelana</i>
ML525916301	Venezuela	Distrito Capital	<i>venezuelana</i>
ML542417511	Venezuela	Falcón	<i>venezuelana</i>
ML542423641	Venezuela	Falcón	<i>venezuelana</i>
ML286940	Venezuela	Falcón	<i>venezuelana</i>
ML62155	Venezuela	Lara	<i>venezuelana</i>
ML700018	Venezuela	Lara	<i>venezuelana</i>
ML37632091	Venezuela	Lara	<i>venezuelana</i>
ML62154	Venezuela	Mérida	<i>venezuelana</i>
ML286726	Venezuela	Mérida	<i>venezuelana</i>
ML324094161	Venezuela	Trujillo	<i>venezuelana</i>
ML582028001	Venezuela	Vargas	<i>venezuelana</i>
ML582027991	Venezuela	Vargas	<i>venezuelana</i>
ML290947	Venezuela	Yaracuy	<i>venezuelana</i>
ML586943581	Venezuela	Yaracuy	<i>venezuelana</i>
ML238422091	Venezuela	Yaracuy	<i>venezuelana</i>

Variables description

1. **lowFreq**: minimum frequency (Hz) of the selection area.
2. **highFreq**: maximum frequency (Hz) of the selection area.
3. **deltaFreq**: frequency range (Hz) of the selection area.
4. **peakFreq**: peak frequency (Hz) of the selection area.
5. **centerFreq**: center frequency (Hz) of the selection area.
6. **centerTimeRel**: Center time relative.
7. **avgEntropy**: Average entropy.
8. **BW50**: Bandwidth 50% (Hz).
9. **deltaTime**: time duration in seconds.
10. **Notes**: number of notes.
11. **notesRate**: number of notes per second (Notes/Delta Time).
12. **meanTimeInterval***: average time between notes' peak frequencies in seconds within a song phrase.
13. **timeIntervalSD***: standard deviation of the time intervals between notes' peak frequency in seconds.
14. **firstTimeInterval***: average time between notes' peak frequencies in seconds within the phrase' first half.
15. **lastTimeInterval***: average time between notes' peak frequencies in seconds within the phrase' last half.
16. **firstMeanFreq***: mean peak frequency (Hz) within the phrase' first half.
17. **lastMeanFreq***: mean peak frequency (Hz) within the phrase' last half.

*A customized R script was developed to generate these variables by calculating the mean of the frequency contour data extracted from Raven Pro (including Peak Frequency Contour and Frequency Contour at 5, 25, 50, 75, and 95%). The script also identifies the temporal position of local peaks, defined as the peak frequency of each song note. Visual inspection was used to validate the positions of the notes' peak frequencies. Using these positions, both their time coordinates and frequency values were determined, which enabled calculation of average time intervals and frequency values. These variables provide a robust estimation of the temporal distribution between notes and the variation in peak frequencies, thereby enhancing metrics such as Notes Rate, Peak Frequency, and Center Frequency.

Table 2. Descriptive statistics of the low frequency (lowFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	582.83 $\pm$ 108.586; 436.12-753.88; n = 14
<i>boliviana</i>	2	360.47 $\pm$ 64.942; 221.2-492.85; n = 14
<i>campanisona</i>	1	733.47 $\pm$ 91.771; 483.05-964.3; n = 67
<i>campanisona</i>	2	461.55 $\pm$ 74.952; 338-702.7; n = 67
<i>columbiana</i>	1	732.27 $\pm$ 72.258; 612.72-861.89; n = 19
<i>columbiana</i>	2	387.35 $\pm$ 88.085; 247.5-563.4; n = 19
<i>fulvescens</i>	1	650.73 $\pm$ 80.463; 485.92-754.01; n = 11
<i>fulvescens</i>	2	409.21 $\pm$ 53.358; 316-502.3; n = 11
North SFR	1	653.94 $\pm$ 82.812; 518.18-827.7; n = 17
North SFR	2	450.72 $\pm$ 58.748; 367.3-570.1; n = 17
<i>obscura</i>	1	488.54 $\pm$ 61.597; 425.3-580.12; n = 9
<i>obscura</i>	2	329.89 $\pm$ 71.738; 235.5-438; n = 9
<i>olivacea/berlepschi</i>	1	583.19 $\pm$ 84.849; 354.33-725.18; n = 16
<i>olivacea/berlepschi</i>	2	302.68 $\pm$ 74.098; 176.68-403.2; n = 16
<i>punctigula</i>	1	718.4 $\pm$ 94.499; 567.4-913.8; n = 31
<i>punctigula</i>	2	363.75 $\pm$ 73.884; 197.8-489.5; n = 31
<i>tshororo</i>	1	705.19 $\pm$ 99.305; 398.67-878.94; n = 49
<i>tshororo</i>	2	454.37 $\pm$ 93.449; 249.2-608.2; n = 49
<i>venezuelana</i>	1	588.85 $\pm$ 75.701; 372.75-726.92; n = 54
<i>venezuelana</i>	2	368.11 $\pm$ 66.527; 218.3-515.6; n = 54

Table 3. Descriptive statistics of the high frequency (highFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	1220.5 $\pm$ 72.012; 1130.75-1336.4; n = 14
<i>boliviana</i>	2	1219.34 $\pm$ 73.32; 1060.1-1318.4; n = 14
<i>campanisona</i>	1	1232.02 $\pm$ 107.66; 1060.66-1681.8; n = 67
<i>campanisona</i>	2	1235.31 $\pm$ 86.845; 1010.9-1506.8; n = 67
<i>columbiana</i>	1	1221.94 $\pm$ 116.114; 1011.64-1382.6; n = 19
<i>columbiana</i>	2	1162.38 $\pm$ 52.535; 1041.9-1248.8; n = 19
<i>fulvescens</i>	1	954.8 $\pm$ 20.754; 915.5-977.41; n = 11
<i>fulvescens</i>	2	958.89 $\pm$ 139.819; 752.79-1139.3; n = 11
North SFR	1	1221.88 $\pm$ 22.243; 1193.08-1272.13; n = 17
North SFR	2	1318.18 $\pm$ 69.168; 1227.7-1447.1; n = 17
<i>obscura</i>	1	982.8 $\pm$ 37.812; 908.8-1030.4; n = 9
<i>obscura</i>	2	867.73 $\pm$ 131.703; 757.7-1103.3; n = 9
<i>olivacea/berlepsi</i>	1	1215.17 $\pm$ 109.909; 1073.35-1436.6; n = 16
<i>olivacea/berlepsi</i>	2	1205.53 $\pm$ 65.875; 1126.35-1322.6; n = 16
<i>punctigula</i>	1	1239.48 $\pm$ 113.763; 1104.21-1501.2; n = 31
<i>punctigula</i>	2	1175.42 $\pm$ 75.291; 983.6-1310.73; n = 31
<i>tshororo</i>	1	1204.62 $\pm$ 54.86; 1123.67-1321.2; n = 49
<i>tshororo</i>	2	1249.55 $\pm$ 69.123; 1010.57-1407.1; n = 49
<i>venezuelana</i>	1	1048.82 $\pm$ 66.035; 940.72-1247.9; n = 54
<i>venezuelana</i>	2	1088.49 $\pm$ 44.056; 1005.6-1200.7; n = 54

Table 4. Descriptive statistics of the delta frequency (deltaFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	637.67 $\pm$ 137.53; 437.19-889.68; n = 14
<i>boliviana</i>	2	858.87 $\pm$ 75.076; 696.6-984; n = 14
<i>campanisona</i>	1	498.55 $\pm$ 123.171; 299.04-953.79; n = 67
<i>campanisona</i>	2	773.76 $\pm$ 112.768; 454.1-1085.9; n = 67
<i>columbiana</i>	1	489.67 $\pm$ 95.485; 357.4-683.5; n = 19
<i>columbiana</i>	2	775.03 $\pm$ 123.948; 478.5-924.73; n = 19
<i>fulvescens</i>	1	304.07 $\pm$ 75.463; 195.69-449.38; n = 11
<i>fulvescens</i>	2	549.68 $\pm$ 179.077; 301.49-784.3; n = 11
North SFR	1	567.93 $\pm$ 87.766; 411.11-696.43; n = 17
North SFR	2	867.45 $\pm$ 67.386; 763.3-993.4; n = 17
<i>obscura</i>	1	494.26 $\pm$ 84.494; 366.98-600.4; n = 9
<i>obscura</i>	2	537.84 $\pm$ 188.487; 349.3-867.8; n = 9
<i>olivacea/berlepsi</i>	1	631.97 $\pm$ 113.638; 477.04-932.47; n = 16
<i>olivacea/berlepsi</i>	2	902.85 $\pm$ 68.683; 792-1012.1; n = 16
<i>punctigula</i>	1	521.08 $\pm$ 114.88; 325.24-722.2; n = 31
<i>punctigula</i>	2	811.67 $\pm$ 69.214; 688.1-959.5; n = 31
<i>tshororo</i>	1	499.44 $\pm$ 109.623; 326.36-857.13; n = 49
<i>tshororo</i>	2	795.18 $\pm$ 107.239; 612.6-1066.4; n = 49
<i>venezuelana</i>	1	459.97 $\pm$ 89.502; 312.09-766.85; n = 54
<i>venezuelana</i>	2	720.38 $\pm$ 74.2; 562.5-890.47; n = 54

Table 5. Descriptive statistics of the peak frequency (peakFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	1124.34 $\pm$ 74.679; 1012.06-1227.39; n = 14
<i>boliviana</i>	2	1055.13 $\pm$ 89.782; 904.4-1184.33; n = 14
<i>campanisona</i>	1	1097.6 $\pm$ 58.813; 960.94-1248.93; n = 67
<i>campanisona</i>	2	1053.4 $\pm$ 100.988; 820.31-1270.46; n = 67
<i>columbiana</i>	1	1030.81 $\pm$ 64.359; 904.4-1141.26; n = 19
<i>columbiana</i>	2	965.41 $\pm$ 60.086; 882.86-1055.13; n = 19
<i>fulvescens</i>	1	842.48 $\pm$ 25.929; 796.73-882.86; n = 11
<i>fulvescens</i>	2	775.06 $\pm$ 55.334; 689.06-904.4; n = 11
North SFR	1	1106.82 $\pm$ 28.87; 1054.69-1148.44; n = 17
North SFR	2	1064.56 $\pm$ 126.908; 843.75-1270.46; n = 17
<i>obscura</i>	1	840.77 $\pm$ 21.467; 796.88-866.32; n = 9
<i>obscura</i>	2	721.04 $\pm$ 36.869; 679.01-796.88; n = 9
<i>olivacea/berlepschi</i>	1	1061.08 $\pm$ 63.513; 947.46-1173.05; n = 16
<i>olivacea/berlepschi</i>	2	980.36 $\pm$ 99.125; 839.8-1125; n = 16
<i>punctigula</i>	1	1070.94 $\pm$ 51.008; 984.38-1162.79; n = 31
<i>punctigula</i>	2	1001.15 $\pm$ 87.951; 843.75-1171.88; n = 31
<i>tshororo</i>	1	1095.35 $\pm$ 42.192; 1012.06-1184.33; n = 49
<i>tshororo</i>	2	1094.93 $\pm$ 88.551; 818.26-1227.39; n = 49
<i>venezuelana</i>	1	939.14 $\pm$ 56.091; 843.75-1076.66; n = 54
<i>venezuelana</i>	2	904.24 $\pm$ 77.397; 725.84-1098.19; n = 54

Table 6. Descriptive statistics of the center frequency (centerFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	1113.57 $\pm$ 64.56; 1033.59-1205.86; n = 14
<i>boliviana</i>	2	1024.37 $\pm$ 101.252; 882.86-1184.33; n = 14
<i>campanisona</i>	1	1092.06 $\pm$ 56.543; 960.94-1227.39; n = 67
<i>campanisona</i>	2	1007.12 $\pm$ 75.123; 843.75-1205.86; n = 67
<i>columbiana</i>	1	1030.71 $\pm$ 60.657; 904.4-1119.73; n = 19
<i>columbiana</i>	2	958.41 $\pm$ 48.883; 882.86-1031.25; n = 19
<i>fulvescens</i>	1	850.49 $\pm$ 27.385; 818.26-904.4; n = 11
<i>fulvescens</i>	2	752.66 $\pm$ 29.235; 689.06-796.73; n = 11
North SFR	1	1097.73 $\pm$ 21.251; 1076.66-1148.44; n = 17
North SFR	2	1015.42 $\pm$ 62.794; 890.62-1098.19; n = 17
<i>obscura</i>	1	825.14 $\pm$ 25.214; 796.88-866.32; n = 9
<i>obscura</i>	2	708.03 $\pm$ 25.842; 679.01-750; n = 9
<i>olivacea/berlepschi</i>	1	1053.99 $\pm$ 62.869; 968.99-1196.51; n = 16
<i>olivacea/berlepschi</i>	2	869.82 $\pm$ 107.601; 656.25-1054.69; n = 16
<i>punctigula</i>	1	1060.85 $\pm$ 50.681; 984.38-1162.79; n = 31
<i>punctigula</i>	2	936.48 $\pm$ 65.353; 796.88-1076.66; n = 31
<i>tshororo</i>	1	1086.48 $\pm$ 41.993; 1012.06-1184.33; n = 49
<i>tshororo</i>	2	1036.94 $\pm$ 80.788; 796.73-1205.86; n = 49
<i>venezuelana</i>	1	931.45 $\pm$ 49.526; 861.33-1055.13; n = 54
<i>venezuelana</i>	2	888.07 $\pm$ 56.634; 749.25-1055.13; n = 54

Table 7. Descriptive statistics of the center time relative (centerTimeRel).

Group	Phrase	Statistics
<i>boliviana</i>	1	0.71 ± 0.043 ; 0.66-0.79; n = 14
<i>boliviana</i>	2	0.33 ± 0.097 ; 0.14-0.49; n = 14
<i>campanisona</i>	1	0.67 ± 0.05 ; 0.5-0.76; n = 67
<i>campanisona</i>	2	0.33 ± 0.134 ; 0.07-0.87; n = 67
<i>columbiana</i>	1	0.73 ± 0.046 ; 0.66-0.8; n = 19
<i>columbiana</i>	2	0.38 ± 0.074 ; 0.24-0.53; n = 19
<i>fulvescens</i>	1	0.65 ± 0.093 ; 0.45-0.75; n = 11
<i>fulvescens</i>	2	0.53 ± 0.226 ; 0.2-0.86; n = 11
North SFR	1	0.62 ± 0.036 ; 0.54-0.69; n = 17
North SFR	2	0.36 ± 0.078 ; 0.23-0.51; n = 17
<i>obscura</i>	1	0.69 ± 0.042 ; 0.65-0.77; n = 9
<i>obscura</i>	2	0.38 ± 0.142 ; 0.21-0.57; n = 9
<i>olivacea/berlepsi</i>	1	0.71 ± 0.074 ; 0.49-0.79; n = 16
<i>olivacea/berlepsi</i>	2	0.36 ± 0.091 ; 0.17-0.47; n = 16
<i>punctigula</i>	1	0.74 ± 0.058 ; 0.59-0.81; n = 31
<i>punctigula</i>	2	0.36 ± 0.144 ; 0.12-0.69; n = 31
<i>tshororo</i>	1	0.68 ± 0.046 ; 0.49-0.76; n = 49
<i>tshororo</i>	2	0.3 ± 0.134 ; 0.12-0.79; n = 49
<i>venezuelana</i>	1	0.74 ± 0.069 ; 0.54-0.85; n = 54
<i>venezuelana</i>	2	0.37 ± 0.094 ; 0.21-0.62; n = 54

Table 8. Descriptive statistics of the average entropy (avgEntropy).

Group	Phrase	Statistics
<i>boliviana</i>	1	2.6 ± 0.15 ; 2.3-2.8; n = 14
<i>boliviana</i>	2	3.6 ± 0.16 ; 3.4-3.9; n = 14
<i>campanisona</i>	1	2.5 ± 0.2 ; 2.1-3; n = 67
<i>campanisona</i>	2	3.7 ± 0.3 ; 3.1-4.5; n = 67
<i>columbiana</i>	1	2.6 ± 0.2 ; 2.3-2.9; n = 19
<i>columbiana</i>	2	3.6 ± 0.3 ; 3.3-4; n = 19
<i>fulvescens</i>	1	2.4 ± 0.1 ; 2.23-2.5; n = 11
<i>fulvescens</i>	2	3.1 ± 0.4 ; 2.6-3.8; n = 11
North SFR	1	2.5 ± 0.1 ; 2.4-2.8; n = 17
North SFR	2	4 ± 0.2 ; 3.8-4.3; n = 17
<i>obscura</i>	1	2.5 ± 0.1 ; 2.4-2.7; n = 9
<i>obscura</i>	2	2.8 ± 0.2 ; 2.4-3; n = 9
<i>olivacea/berlepsi</i>	1	2.8 ± 0.3 ; 2.2-3.4; n = 16
<i>olivacea/berlepsi</i>	2	3.5 ± 0.4 ; 2.7-4; n = 16
<i>punctigula</i>	1	2.5 ± 0.2 ; 2.2-2.9; n = 31
<i>punctigula</i>	2	3.5 ± 0.35 ; 2.8-4.3; n = 31
<i>tshororo</i>	1	2.4 ± 0.15 ; 2.2-2.7; n = 49
<i>tshororo</i>	2	3.8 ± 0.3 ; 3-4.3; n = 49
<i>venezuelana</i>	1	2.6 ± 0.2 ; 2.3-3; n = 54
<i>venezuelana</i>	2	3.4 ± 0.3 ; 2.7-4.1; n = 54

Table 9. Descriptive statistics of the bandwidth 50% (BW50).

Group	Phrase	Statistics
<i>boliviana</i>	1	41.5 ± 17.8; 21.5-64.6; n = 14
<i>boliviana</i>	2	103.1 ± 44.0; 43.1-172.3; n = 14
<i>campanisona</i>	1	42.5 ± 15.7; 21.5-93.8; n = 67
<i>campanisona</i>	2	155.2 ± 49.9; 43.1-323; n = 67
<i>columbiana</i>	1	38.2 ± 9.9; 21.5-46.9; n = 19
<i>columbiana</i>	2	125.9 ± 44.5; 64.6-210.9; n = 19
<i>fulvescens</i>	1	40.2 ± 9.4; 21.5-46.9; n = 11
<i>fulvescens</i>	2	75.9 ± 29; 43.1-129.2; n = 11
North SFR	1	42.3 ± 11.7; 21.5-70.3; n = 17
North SFR	2	205.8 ± 48.9; 117.2-281.3; n = 17
<i>obscura</i>	1	41.7 ± 10.3; 23.4-46.9; n = 9
<i>obscura</i>	2	57.3 ± 26.5; 23.4-93.8; n = 9
<i>olivacea/berlepsi</i>	1	64.4 ± 36.5; 21.5-140.8; n = 16
<i>olivacea/berlepsi</i>	2	232.2 ± 119.9; 117.2-468.8; n = 16
<i>punctigula</i>	1	39.4 ± 14.3; 21.5-93.8; n = 31
<i>punctigula</i>	2	129.6 ± 32; 64.6-193.8; n = 31
<i>tshororo</i>	1	38.5 ± 10.8; 21.5-64.6; n = 49
<i>tshororo</i>	2	166.9 ± 94.7; 86.1-581.4; n = 49
<i>venezuelana</i>	1	37.4 ± 11.7; 15.6-64.6; n = 54
<i>venezuelana</i>	2	130.5 ± 37.2; 62.5-193.8; n = 54

Table 10. Descriptive statistics of the delta time (deltaTime).

Group	Phrase	Statistics
<i>boliviana</i>	1	6.78 ± 0.378 ; 6.29-7.45; n = 14
<i>boliviana</i>	2	2.68 ± 1.012 ; 0.86-4.25; n = 14
<i>campanisona</i>	1	9.88 ± 2.212 ; 7.28-19.66; n = 67
<i>campanisona</i>	2	1.68 ± 1.046 ; 0.07-5.35; n = 67
<i>columbiana</i>	1	5.9 ± 0.975 ; 2.68-7.56; n = 19
<i>columbiana</i>	2	2.87 ± 0.961 ; 0.87-5.37; n = 19
<i>fulvescens</i>	1	7.95 ± 1.636 ; 5.88-11.42; n = 11
<i>fulvescens</i>	2	0.85 ± 0.528 ; 0.17-1.89; n = 11
North SFR	1	9.71 ± 1.191 ; 7.82-11.95; n = 17
North SFR	2	3.92 ± 1.238 ; 1.72-5.88; n = 17
<i>obscura</i>	1	7.28 ± 1.428 ; 5.64-9.92; n = 9
<i>obscura</i>	2	1 ± 0.668 ; 0.14-1.75; n = 9
<i>olivacea/berlepschi</i>	1	8.2 ± 1.075 ; 6.3-9.59; n = 16
<i>olivacea/berlepschi</i>	2	4.19 ± 0.85 ; 2.94-5.41; n = 16
<i>punctigula</i>	1	5.35 ± 0.714 ; 3.45-6.6; n = 31
<i>punctigula</i>	2	2.83 ± 1.176 ; 0.19-4.71; n = 31
<i>tshororo</i>	1	9.7 ± 1.904 ; 5.37-16.58; n = 49
<i>tshororo</i>	2	0.91 ± 0.435 ; 0.21-2.01; n = 49
<i>venezuelana</i>	1	4.62 ± 0.552 ; 3.46-5.56; n = 54
<i>venezuelana</i>	2	3.05 ± 0.575 ; 1.75-5.09; n = 54

Table 11. Descriptive statistics of the number of notes.

Group	Phrase	Statistics
<i>boliviana</i>	1	54.5 ± 4.451 ; 48-64; n = 14
<i>boliviana</i>	2	12.07 ± 5.03 ; 4-21; n = 14
<i>campanisona</i>	1	61.99 ± 14.627 ; 42-130; n = 67
<i>campanisona</i>	2	12.46 ± 7.917 ; 1-37; n = 67
<i>columbiana</i>	1	25.47 ± 4.477 ; 11-32; n = 19
<i>columbiana</i>	2	8.26 ± 2.746 ; 3-14; n = 19
<i>fulvescens</i>	1	53.18 ± 8.841 ; 36-64; n = 11
<i>fulvescens</i>	2	2.64 ± 1.286 ; 1-5; n = 11
North SFR	1	94.24 ± 14.754 ; 72-113; n = 17
North SFR	2	40.88 ± 14.035 ; 17-66; n = 17
<i>obscura</i>	1	53.56 ± 8.487 ; 44-67; n = 9
<i>obscura</i>	2	3.33 ± 1.803 ; 1-5; n = 9
<i>olivacea/berlepschi</i>	1	61.88 ± 9.892 ; 47-76; n = 16
<i>olivacea/berlepschi</i>	2	23.94 ± 5.721 ; 14-35; n = 16
<i>punctigula</i>	1	23.13 ± 3.222 ; 15-29; n = 31
<i>punctigula</i>	2	8.48 ± 3.213 ; 1-14; n = 31
<i>tshororo</i>	1	70.22 ± 15.866 ; 42-127; n = 49
<i>tshororo</i>	2	6.51 ± 2.945 ; 2-13; n = 49
<i>venezuelana</i>	1	15.5 ± 1.587 ; 12-18; n = 54
<i>venezuelana</i>	2	8.52 ± 1.397 ; 5-13; n = 54

Table 12. Descriptive statistics of the notes rate (notesRate).

Group	Phrase	Statistics
<i>boliviana</i>	1	8.06 ± 0.688 ; 7.08-9.05; n = 14
<i>boliviana</i>	2	4.47 ± 0.349 ; 3.86-4.94; n = 14
<i>campanisona</i>	1	6.28 ± 0.61 ; 4.97-7.21; n = 67
<i>campanisona</i>	2	7.52 ± 1.268 ; 5.94-14.45; n = 67
<i>columbiana</i>	1	4.32 ± 0.338 ; 3.43-4.79; n = 19
<i>columbiana</i>	2	2.9 ± 0.301 ; 2.6-3.47; n = 19
<i>fulvescens</i>	1	6.8 ± 1.106 ; 4.99-8.28; n = 11
<i>fulvescens</i>	2	3.57 ± 1.117 ; 2.64-5.93; n = 11
North SFR	1	9.69 ± 0.774 ; 8.25-10.9; n = 17
North SFR	2	10.34 ± 0.643 ; 8.93-11.51; n = 17
<i>obscura</i>	1	7.43 ± 0.696 ; 5.93-8.06; n = 9
<i>obscura</i>	2	4.12 ± 1.598 ; 2.85-7.1; n = 9
<i>olivacea/berlepschi</i>	1	7.52 ± 0.337 ; 7.04-7.96; n = 16
<i>olivacea/berlepschi</i>	2	5.78 ± 1.202 ; 4.07-7.19; n = 16
<i>punctigula</i>	1	4.34 ± 0.436 ; 3.61-5.19; n = 31
<i>punctigula</i>	2	3.16 ± 0.581 ; 2.54-5.26; n = 31
<i>tshororo</i>	1	7.22 ± 0.482 ; 6.39-8.07; n = 49
<i>tshororo</i>	2	7.35 ± 0.666 ; 6.21-9.32; n = 49
<i>venezuelana</i>	1	3.37 ± 0.235 ; 2.7-3.87; n = 54
<i>venezuelana</i>	2	2.81 ± 0.173 ; 2.51-3.39; n = 54

Table 13. Descriptive statistics of the mean time interval (meanTimeInterval).

Group	Phrase	Statistics
<i>boliviana</i>	1	0.13 ± 0.011 ; 0.11-0.14; n = 14
<i>boliviana</i>	2	0.24 ± 0.022 ; 0.21-0.28; n = 14
<i>campanisona</i>	1	0.16 ± 0.016 ; 0.14-0.2; n = 67
<i>campanisona</i>	2	0.14 ± 0.035 ; 0-0.18; n = 67
<i>columbiana</i>	1	0.24 ± 0.021 ; 0.21-0.3; n = 19
<i>columbiana</i>	2	0.38 ± 0.038 ; 0.3-0.43; n = 19
<i>fulvescens</i>	1	0.15 ± 0.027 ; 0.12-0.2; n = 11
<i>fulvescens</i>	2	0.32 ± 0.166 ; 0-0.47; n = 11
North SFR	1	0.1 ± 0.008 ; 0.09-0.12; n = 17
North SFR	2	0.1 ± 0.007 ; 0.09-0.12; n = 17
<i>obscura</i>	1	0.14 ± 0.014 ; 0.12-0.17; n = 9
<i>obscura</i>	2	0.29 ± 0.169 ; 0-0.41; n = 9
<i>olivacea/berlepschi</i>	1	0.13 ± 0.006 ; 0.12-0.14; n = 16
<i>olivacea/berlepschi</i>	2	0.18 ± 0.04 ; 0.14-0.26; n = 16
<i>punctigula</i>	1	0.24 ± 0.03 ; 0.19-0.32; n = 31
<i>punctigula</i>	2	0.34 ± 0.097 ; 0-0.42; n = 31
<i>tshororo</i>	1	0.14 ± 0.009 ; 0.12-0.16; n = 49
<i>tshororo</i>	2	0.15 ± 0.015 ; 0.12-0.2; n = 49
<i>venezuelana</i>	1	0.31 ± 0.023 ; 0.27-0.39; n = 54
<i>venezuelana</i>	2	0.39 ± 0.025 ; 0.31-0.44; n = 54

Table 14. Descriptive statistics of the time interval standard deviation (timeIntervalSD).

Group	Phrase	Statistics
<i>boliviana</i>	1	0.03 ± 0.019 ; 0.02-0.09; n = 14
<i>boliviana</i>	2	0.01 ± 0.006 ; 0-0.03; n = 14
<i>campanisona</i>	1	0.02 ± 0.008 ; 0.01-0.06; n = 67
<i>campanisona</i>	2	0.02 ± 0.039 ; 0-0.18; n = 67
<i>columbiana</i>	1	0.05 ± 0.019 ; 0.02-0.09; n = 19
<i>columbiana</i>	2	0.02 ± 0.019 ; 0.01-0.1; n = 19
<i>fulvescens</i>	1	0.12 ± 0.035 ; 0.08-0.18; n = 11
<i>fulvescens</i>	2	0.16 ± 0.195 ; 0-0.47; n = 11
North SFR	1	0.02 ± 0.003 ; 0.01-0.02; n = 17
North SFR	2	0.02 ± 0.04 ; 0.01-0.18; n = 17
<i>obscura</i>	1	0.14 ± 0.067 ; 0.09-0.3; n = 9
<i>obscura</i>	2	0.12 ± 0.173 ; 0-0.41; n = 9
<i>olivacea/berlepsi</i>	1	0.02 ± 0.005 ; 0.01-0.03; n = 16
<i>olivacea/berlepsi</i>	2	0.01 ± 0.007 ; 0.01-0.03; n = 16
<i>punctigula</i>	1	0.06 ± 0.02 ; 0.03-0.11; n = 31
<i>punctigula</i>	2	0.02 ± 0.023 ; 0-0.13; n = 31
<i>tshororo</i>	1	0.03 ± 0.072 ; 0.01-0.53; n = 49
<i>tshororo</i>	2	0.03 ± 0.048 ; 0-0.17; n = 49
<i>venezuelana</i>	1	0.07 ± 0.015 ; 0.04-0.11; n = 54
<i>venezuelana</i>	2	0.03 ± 0.023 ; 0.01-0.16; n = 54

Table 15. Descriptive statistics of the mean time interval within the first half of the phrase (firstTimeInterval).

Group	Phrase	Statistics
<i>boliviana</i>	1	0.14 ± 0.012 ; 0.12-0.17; n = 14
<i>boliviana</i>	2	0.23 ± 0.026 ; 0.2-0.28; n = 14
<i>campanisona</i>	1	0.17 ± 0.02 ; 0.14-0.22; n = 67
<i>campanisona</i>	2	0.14 ± 0.035 ; 0-0.18; n = 67
<i>columbiana</i>	1	0.27 ± 0.028 ; 0.23-0.35; n = 19
<i>columbiana</i>	2	0.37 ± 0.04 ; 0.3-0.42; n = 19
<i>fulvescens</i>	1	0.22 ± 0.046 ; 0.17-0.31; n = 11
<i>fulvescens</i>	2	0.31 ± 0.168 ; 0-0.47; n = 11
North SFR	1	0.11 ± 0.01 ; 0.1-0.13; n = 17
North SFR	2	0.1 ± 0.005 ; 0.09-0.11; n = 17
<i>obscura</i>	1	0.2 ± 0.022 ; 0.18-0.24; n = 9
<i>obscura</i>	2	0.28 ± 0.172 ; 0-0.41; n = 9
<i>olivacea/berlepsi</i>	1	0.15 ± 0.01 ; 0.13-0.16; n = 16
<i>olivacea/berlepsi</i>	2	0.18 ± 0.038 ; 0.14-0.25; n = 16
<i>punctigula</i>	1	0.27 ± 0.039 ; 0.21-0.38; n = 31
<i>punctigula</i>	2	0.33 ± 0.096 ; 0-0.41; n = 31
<i>tshororo</i>	1	0.15 ± 0.01 ; 0.13-0.17; n = 49
<i>tshororo</i>	2	0.15 ± 0.014 ; 0.12-0.2; n = 49
<i>venezuelana</i>	1	0.36 ± 0.03 ; 0.3-0.45; n = 54
<i>venezuelana</i>	2	0.38 ± 0.024 ; 0.31-0.43; n = 54

Table 16. Descriptive statistics of the mean time interval within the last half of the phrase (lastTimeInterval).

Group	Phrase	Statistics
<i>boliviana</i>	1	0.11 ± 0.011 ; 0.1-0.13; n = 14
<i>boliviana</i>	2	0.24 ± 0.019 ; 0.22-0.28; n = 14
<i>campanisona</i>	1	0.15 ± 0.013 ; 0.12-0.18; n = 67
<i>campanisona</i>	2	0.14 ± 0.036 ; 0-0.18; n = 67
<i>columbiana</i>	1	0.2 ± 0.019 ; 0.18-0.25; n = 19
<i>columbiana</i>	2	0.38 ± 0.04 ; 0.29-0.42; n = 19
<i>fulvescens</i>	1	0.08 ± 0.014 ; 0.07-0.11; n = 11
<i>fulvescens</i>	2	0.33 ± 0.168 ; 0-0.47; n = 11
North SFR	1	0.1 ± 0.008 ; 0.08-0.11; n = 17
North SFR	2	0.1 ± 0.009 ; 0.09-0.12; n = 17
<i>obscura</i>	1	0.07 ± 0.006 ; 0.07-0.09; n = 9
<i>obscura</i>	2	0.29 ± 0.168 ; 0-0.41; n = 9
<i>olivacea/berlepsi</i>	1	0.12 ± 0.005 ; 0.11-0.13; n = 16
<i>olivacea/berlepsi</i>	2	0.19 ± 0.042 ; 0.14-0.26; n = 16
<i>punctigula</i>	1	0.21 ± 0.023 ; 0.17-0.27; n = 31
<i>punctigula</i>	2	0.35 ± 0.099 ; 0-0.43; n = 31
<i>tshororo</i>	1	0.13 ± 0.009 ; 0.12-0.15; n = 49
<i>tshororo</i>	2	0.15 ± 0.019 ; 0.1-0.2; n = 49
<i>venezuelana</i>	1	0.26 ± 0.02 ; 0.23-0.33; n = 54
<i>venezuelana</i>	2	0.39 ± 0.028 ; 0.29-0.45; n = 54

Table 17. Descriptive statistics of the mean peak frequency within the first half of the phrase (firstMeanFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	1073.6 $\pm$ 65.095; 984.6-1183.08; n = 14
<i>boliviana</i>	2	1077.89 $\pm$ 83.138; 975.13-1222.45; n = 14
<i>campanisona</i>	1	1081.82 $\pm$ 55.599; 944.87-1193.22; n = 67
<i>campanisona</i>	2	1054.71 $\pm$ 90.506; 800.32-1231.38; n = 67
<i>columbiana</i>	1	1030.18 $\pm$ 56.26; 904.95-1092.01; n = 19
<i>columbiana</i>	2	1042.74 $\pm$ 45.907; 928.62-1115.12; n = 19
<i>fulvescens</i>	1	861.84 $\pm$ 23.146; 833.18-901.41; n = 11
<i>fulvescens</i>	2	795.07 $\pm$ 54.992; 696.24-894.62; n = 11
North SFR	1	1085.53 $\pm$ 23.221; 1044.15-1121.23; n = 17
North SFR	2	1106.07 $\pm$ 69.789; 965.51-1236.82; n = 17
<i>obscura</i>	1	833.98 $\pm$ 26.686; 789.56-863.77; n = 9
<i>obscura</i>	2	745.84 $\pm$ 43.702; 706.32-823.37; n = 9
<i>olivacea/berlepschi</i>	1	1015.67 $\pm$ 58.701; 957.22-1163.62; n = 16
<i>olivacea/berlepschi</i>	2	979.49 $\pm$ 65.751; 874.59-1068.85; n = 16
<i>punctigula</i>	1	1053.71 $\pm$ 64.769; 972.56-1180.23; n = 31
<i>punctigula</i>	2	1006.41 $\pm$ 95.833; 838.43-1196.89; n = 31
<i>tshororo</i>	1	1073.62 $\pm$ 44.047; 991.61-1175.04; n = 49
<i>tshororo</i>	2	1085.78 $\pm$ 74.153; 881.13-1205.24; n = 49
<i>venezuelana</i>	1	914.54 $\pm$ 50.32; 823.56-1021.29; n = 54
<i>venezuelana</i>	2	969.42 $\pm$ 50.01; 836.95-1080.05; n = 54

Table 18. Descriptive statistics of the mean peak frequency within the last half of the phrase (lastMeanFreq).

Group	Phrase	Statistics
<i>boliviana</i>	1	1131.43 $\pm$ 64.63; 1047.38-1231.76; n = 14
<i>boliviana</i>	2	926.36 $\pm$ 134.447; 766.04-1188.63; n = 14
<i>campanisona</i>	1	1106.13 $\pm$ 60.912; 964.1-1228.17; n = 67
<i>campanisona</i>	2	932.29 $\pm$ 93.821; 776.04-1193.51; n = 67
<i>columbiana</i>	1	1049.13 $\pm$ 60.33; 935.87-1148.72; n = 19
<i>columbiana</i>	2	892.31 $\pm$ 68.558; 781.96-1050.19; n = 19
<i>fulvescens</i>	1	844.19 $\pm$ 27.169; 812.44-896.02; n = 11
<i>fulvescens</i>	2	758.54 $\pm$ 46.205; 696.24-862.63; n = 11
North SFR	1	1114.03 $\pm$ 19.415; 1084.41-1152.34; n = 17
North SFR	2	933.2 $\pm$ 113.507; 801.36-1148.98; n = 17
<i>obscura</i>	1	833.07 $\pm$ 15.671; 806.27-852.75; n = 9
<i>obscura</i>	2	702.85 $\pm$ 44.072; 624.16-791.83; n = 9
<i>olivacea/berlepsi</i>	1	1079.05 $\pm$ 78.062; 985.27-1271.75; n = 16
<i>olivacea/berlepsi</i>	2	813.52 $\pm$ 70.203; 740.18-993.97; n = 16
<i>punctigula</i>	1	1086.63 $\pm$ 62.128; 990.58-1198.85; n = 31
<i>punctigula</i>	2	884.43 $\pm$ 102.601; 736.02-1139.44; n = 31
<i>tshororo</i>	1	1098.43 $\pm$ 43.039; 1015.88-1201.14; n = 49
<i>tshororo</i>	2	952.64 $\pm$ 108.452; 637.59-1202.27; n = 49
<i>venezuelana</i>	1	947.51 $\pm$ 50.86; 865.65-1083.51; n = 54
<i>venezuelana</i>	2	809.21 $\pm$ 65.776; 662.96-967.24; n = 54

Table 19. Correlation of the bioacoustics variables along the principal components (PC) selected. To be used for a biological interpretation of the PCs. The suffix P1 or P2 refers to the first and second phrases, respectively.

Variable	PC1	PC2	PC3	PC4
lowFreq.P1	0.48	0.21	-0.58	0.17
highFreq.P1	0.76	0.35	0.15	-0.25
deltaFreq.P1	0.35	0.18	0.67	-0.41
peakFreq.P1	0.88	0.33	0.04	-0.11
centerFreq.P1	0.88	0.33	0.00	-0.07
centerTimeRel.P1	-0.40	0.30	0.04	-0.31
avgEntropy.P1	-0.05	0.31	0.07	0.60
BW50.P1	0.19	0.07	0.43	0.22
lowFreq.P2	0.42	-0.12	-0.67	0.14
highFreq.P2	0.82	0.41	0.00	0.00
deltaFreq.P2	0.54	0.49	0.46	-0.10
peakFreq.P2	0.79	0.35	-0.14	-0.12
centerFreq.P2	0.78	0.38	-0.25	-0.14
centerTimeRel.P2	-0.26	-0.06	-0.25	0.24
avgEntropy.P2	0.60	0.28	-0.27	0.46
BW50.P2	0.47	0.21	0.13	0.43
deltaTime.P1	0.71	-0.45	-0.03	-0.06
deltaTime.P2	-0.17	0.42	0.71	0.33
notes.P1	0.76	-0.54	0.14	0.01
notes.P2	0.39	0.02	0.63	0.37
notesRate.P1	0.71	-0.56	0.26	0.06
notesRate.P2	0.83	-0.37	-0.02	0.08
firstMeanFreq.P1	0.88	0.28	-0.04	-0.11
lastMeanFreq.P1	0.87	0.37	0.04	-0.07
firstMeanFreq.P2	0.75	0.50	-0.18	-0.02
lastMeanFreq.P2	0.65	0.29	-0.45	-0.06
firstTimeInterval.P1	-0.82	0.46	-0.13	0.01
lastTimeInterval.P1	-0.60	0.66	-0.20	0.03
firstTimeInterval.P2	-0.84	0.37	0.05	-0.02
lastTimeInterval.P2	-0.84	0.38	0.06	-0.03
timeIntervalSD.P1	-0.72	-0.16	-0.02	-0.11
timeIntervalSD.P2	-0.19	-0.47	-0.15	0.01
meanTimeInterval.P1	-0.75	0.56	-0.16	0.02
meanTimeInterval.P2	-0.84	0.38	0.06	-0.03

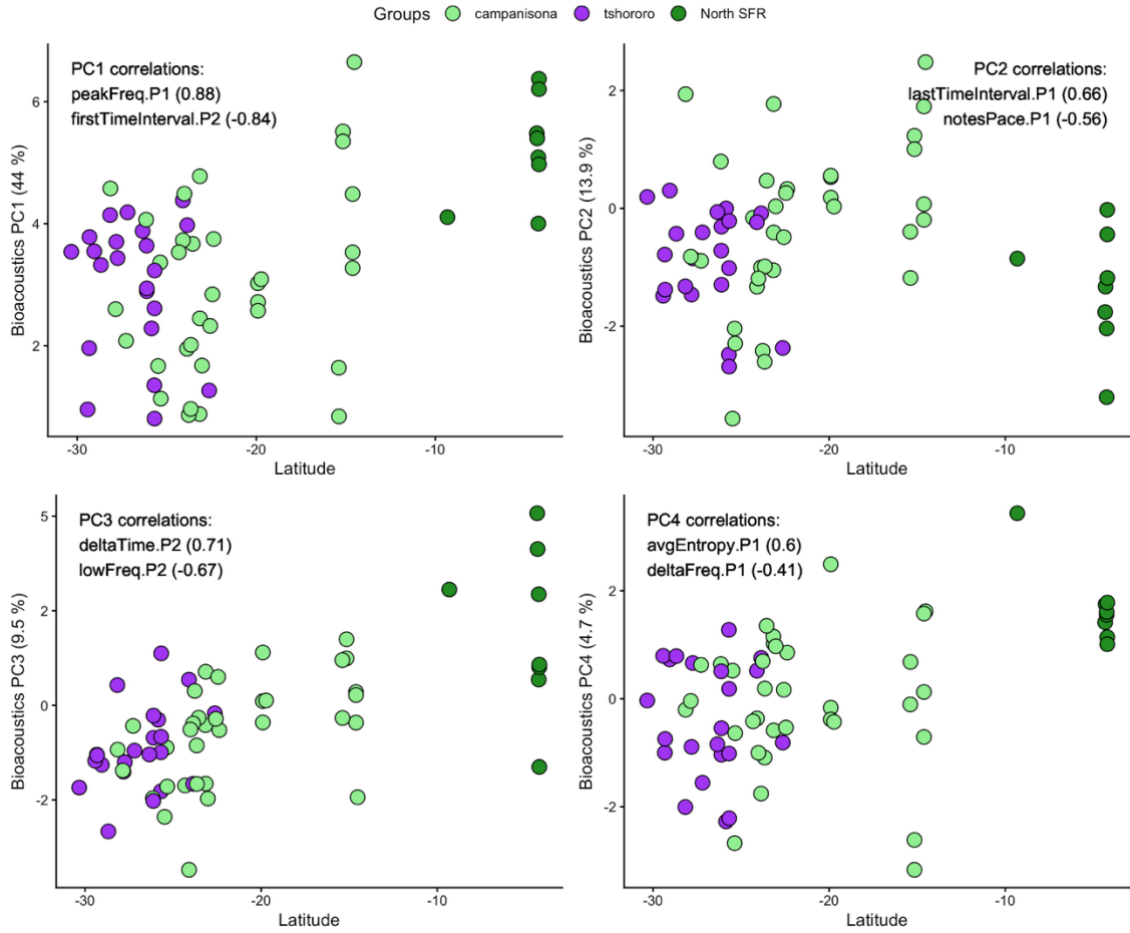


Figure 1. Latitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the Atlantic Forest (*campanisona*, *tshororo*, North SFR). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.

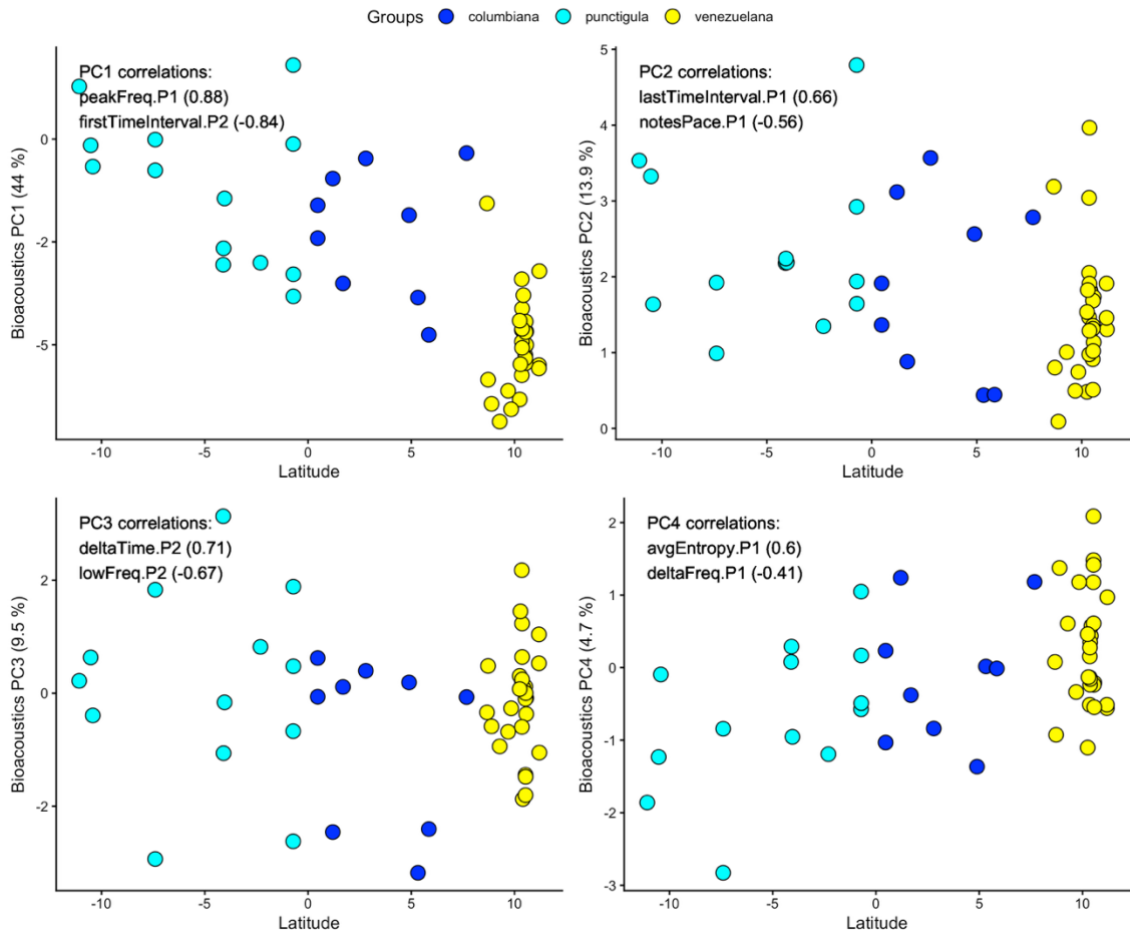


Figure 2. Latitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the north Andes (*columbiana*, *punctigula*, *venezuelana*). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.

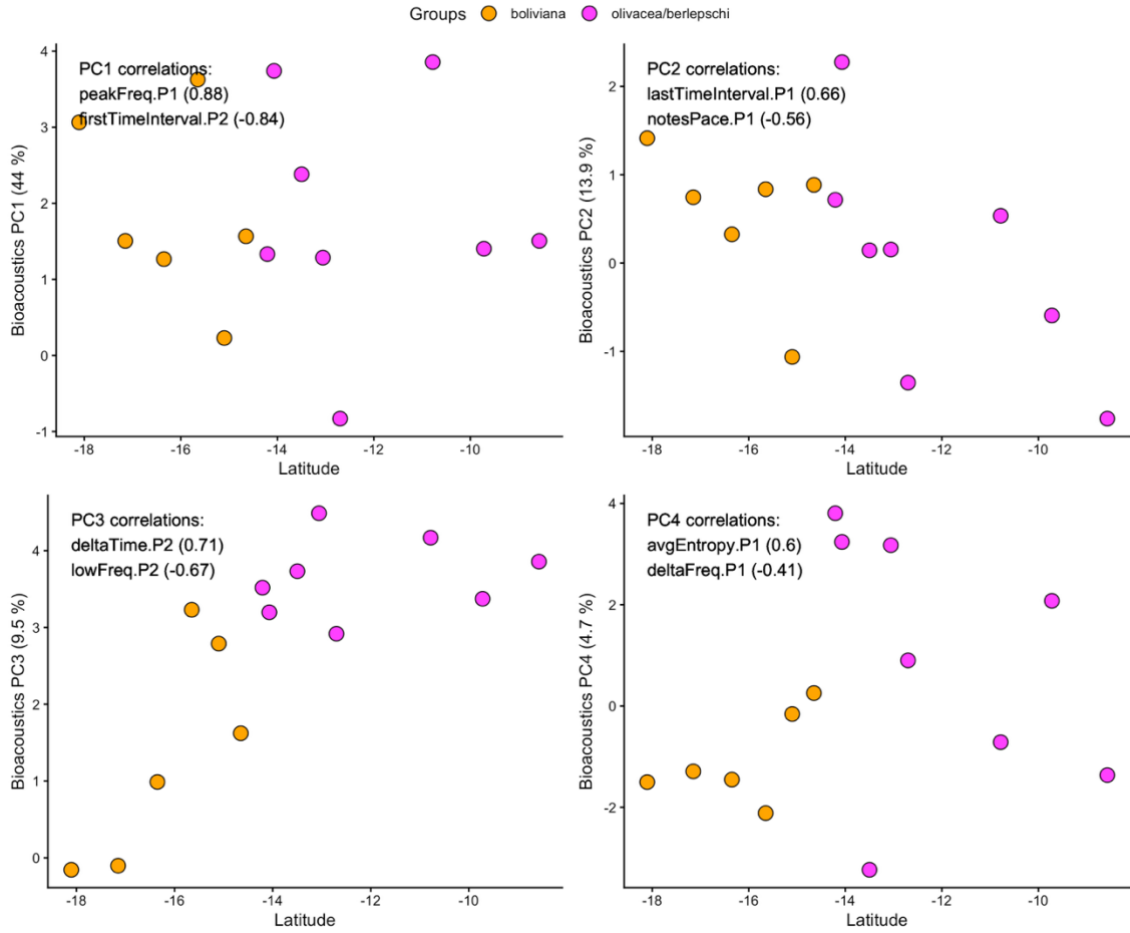


Figure 3. Latitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the south Andes (*boliviana* and *olivacea/berlepschi*). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.

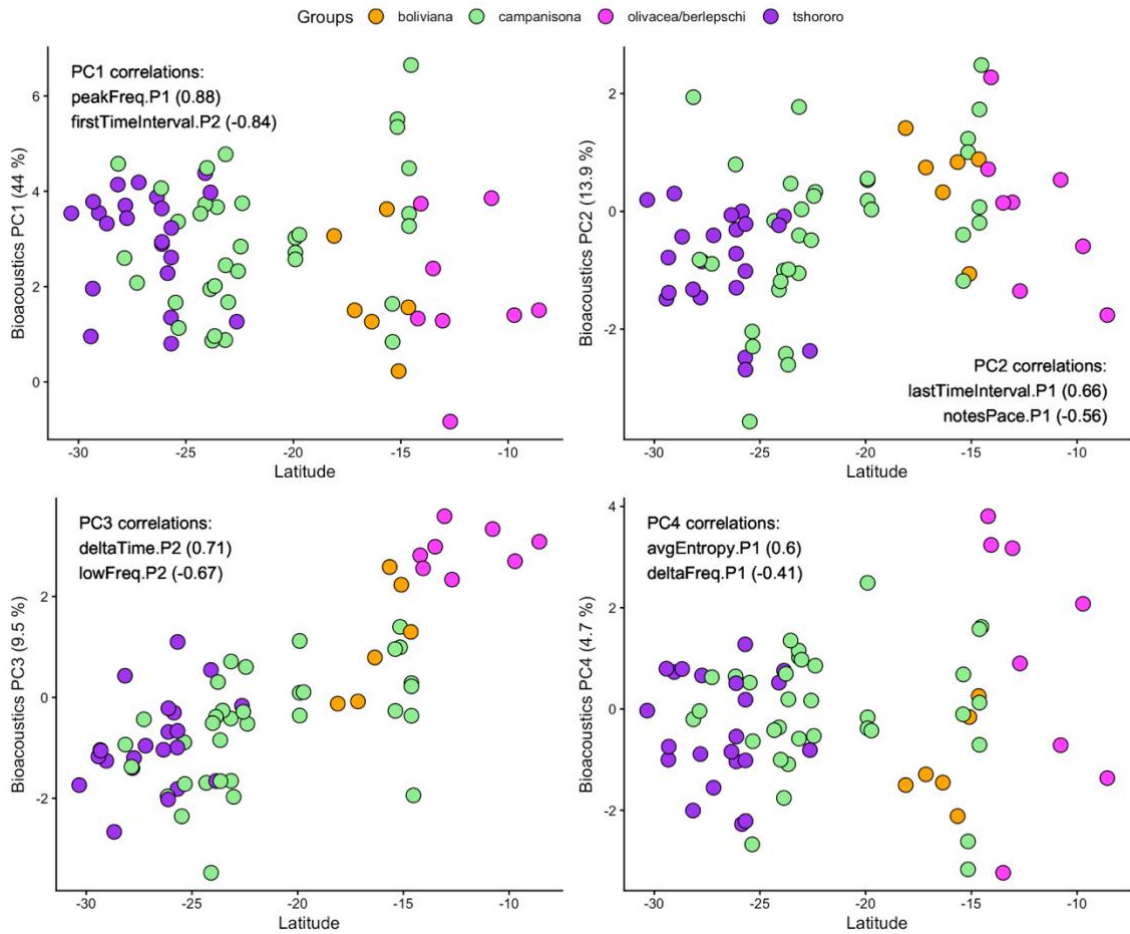


Figure 4. Latitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the south Atlantic Forest (*campanisona* and *tshororo*) and south Andes (*boliviana* and *olivacea/berlepschi*). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.

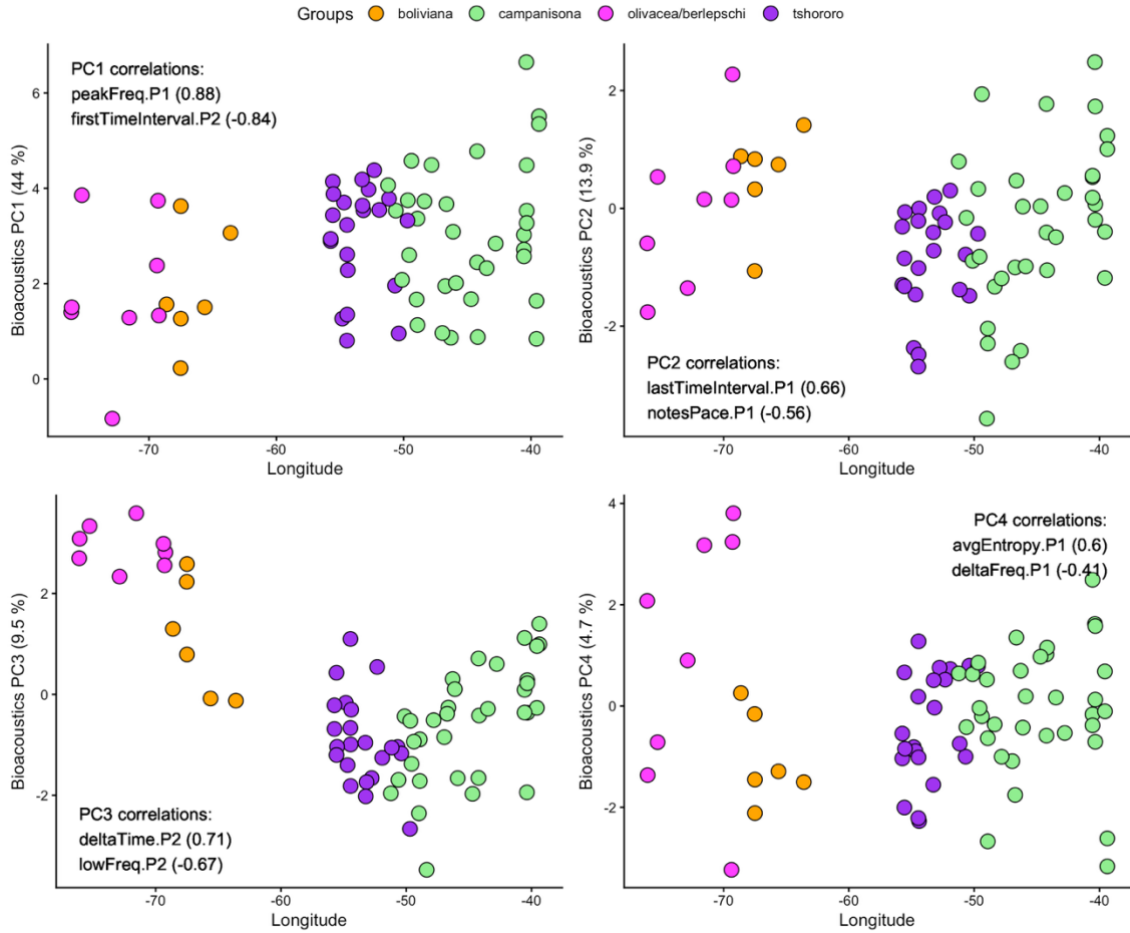


Figure 5. Longitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the south Atlantic Forest (*campanisona* and *tshororo*) and south Andes (*boliviana* and *olivacea/berlepschi*). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.

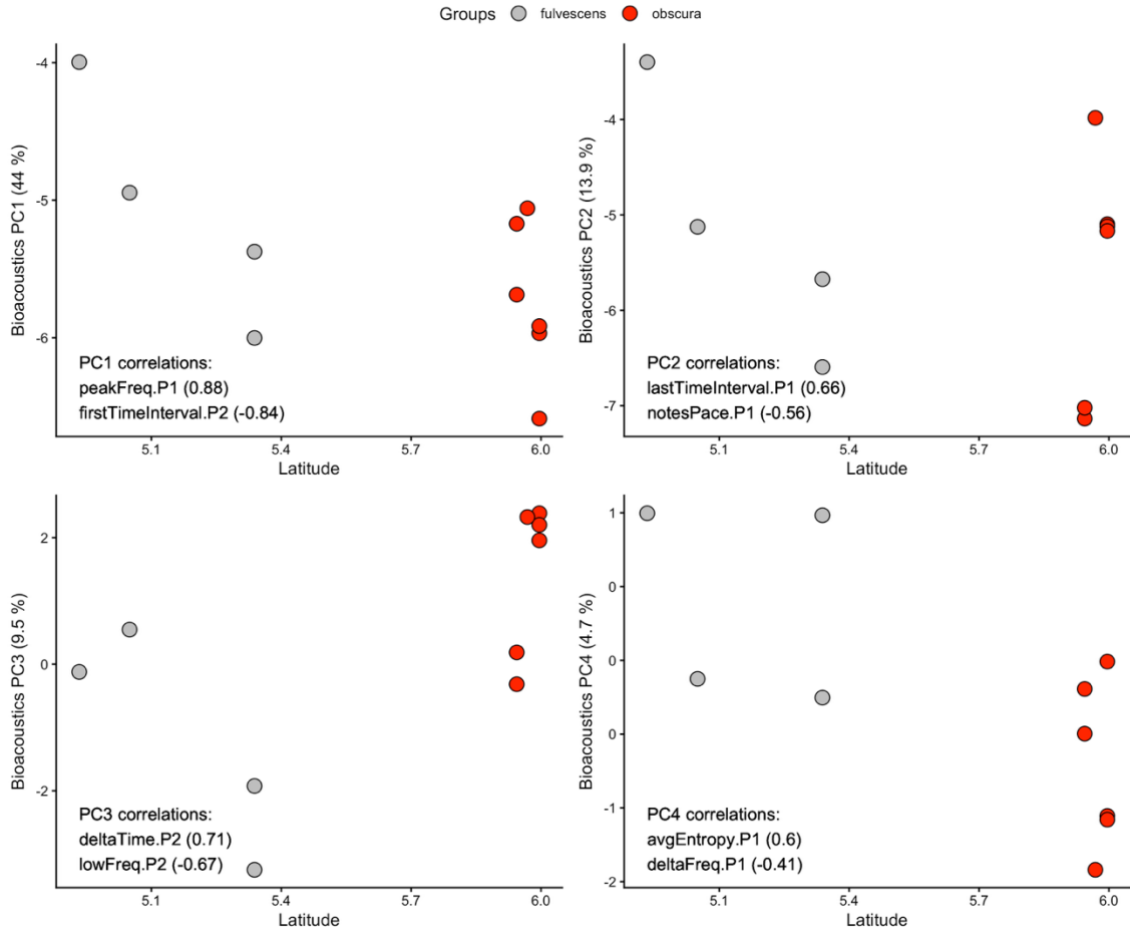


Figure 6. Latitudinal variation in the bioacoustic phenotype (i.e., selected principal components, PC) of *C. campanisona* groups in the Tepuis (*fulvescens* and *obscura*). For biological interpretation, the strongest correlations between each PC and the bioacoustic variables are presented; positive and negative correlations indicate the direction of latitudinal bioacoustic variation. See Table 19 for additional correlations. Variable suffixes P1 and P2 refer to the first and second phrases of the *C. campanisona* song, respectively.